Python Notes: Temperature Conversion Program 🌡️

Celsius ↔ Fahrenheit Using `if-elif-else`

Video: *Python temperature conversion program* 🌡️
Video URL: https://www.youtube.com/watch?v=fy5QsDJmctQ
Level: *Beginner*
Prerequisites: *Variables,* `input()`*, type casting, arithmetic operators, conditional statements,* `round()`*, and f-strings.*

1. Introduction

A ***temperature conversion program*** *converts a temperature from one measurement unit to another. In this project, the user selects the current temperature unit and enters a temperature. Python then uses the appropriate formula to convert it.*

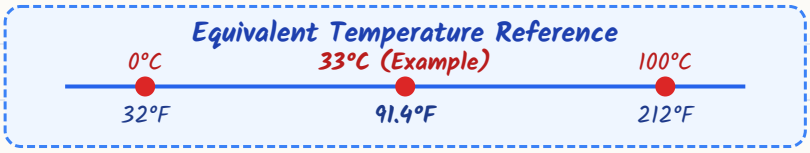
The program supports two units:

- ***Celsius (°C)***
- ***Fahrenheit (°F)***

Example

If the temperature is $33^\circ C$, the equivalent temperature in Fahrenheit is $91.4^\circ F$.

Similarly, if the temperature is $91.4^\circ F$, the equivalent temperature in Celsius is $33^\circ C$.



What you will learn

1. *Accepting user input using* `input()`.
2. *Converting strings into floating-point numbers using* `float()`.
3. *Selecting an operation using* `if`, `elif`, *and* `else`.
4. *Applying mathematical formulas in Python.*
5. *Rounding numerical results with* `round()`.
6. *Displaying results using f-strings.*
7. *Handling invalid unit selections.*

2. Understand the Temperature Conversion Formulas

Before writing the program, understand the two mathematical formulas.

2.1 Celsius to Fahrenheit

To convert Celsius into Fahrenheit, multiply the Celsius temperature by $9/5$, then add 32.

Formula:

$$F = \frac{9}{5}C + 32$$

Example: Convert 33°C to Fahrenheit

Substitute $C = 33$:

$$F = \frac{9}{5}(33) + 32$$

$$F = 59.4 + 32$$

$$F = 91.4^\circ F$$

Python expression:

```
fahrenheit = (celsius * 9 / 5) + 32
```

2.2 Fahrenheit to Celsius

To convert Fahrenheit into Celsius, subtract 32 from the Fahrenheit temperature, then multiply by $5/9$.

Formula:

$$C = (F - 32) \times \frac{5}{9}$$

Where:

- C = temperature in Celsius.
- F = temperature in Fahrenheit.

Example: Convert 91.4°F to Celsius

Substitute $F = 91.4$:

$$C = (91.4 - 32) \times \frac{5}{9}$$

$$C = 59.4 \times \frac{5}{9}$$

$$C = 33^\circ C$$

Python expression:

```
celsius = (fahrenheit - 32) * 5 / 9
```

Quick reference table

Conversion	Formula	Example
Celsius → Fahrenheit	`(C * 9 / 5) + 32`	33°C = 91.4°F
Fahrenheit → Celsius	`(F - 32) * 5 / 9`	91.4°F = 33°C

MOST IMPORTANT

Parentheses matter in the Fahrenheit-to-Celsius formula. Subtract 32 first, then multiply by $5/9$.

3. Step 1: Ask for the Temperature Unit

First, ask the user whether the current temperature is measured in Celsius or Fahrenheit.

4. Step 2: Accept the Temperature

Next, ask the user to enter the temperature.

```
temp = input("Enter the temperature: ")
```

However, `input()` always returns a string—even if the user enters a number.

For example, if the user types `33`, the value is stored as `"33"`, not the number `33`.

Because our program must perform arithmetic, convert the input into a numeric data type.

Correct approach: `float()`

```
temp = float(input("Enter the temperature: "))
```

Explanation

- `input()` accepts the user's entry.
- `float()` converts a valid numeric string into a floating-point number.
- The converted value is assigned to `temp`.

Example:

```
temp = float("33")

print(temp)
print(type(temp))
```

Output:

```
33.0
<class 'float'>
```

Why use `float()` instead of `int()`?

Temperatures can include decimal values, such as:

- `36.5`
- `98.6`
- `-2.7`

Using `int()` would reject decimal input such as `"36.5"`. Using `float()` allows both whole numbers and decimal numbers.

KEY CONCEPT

Use `float(input( ... ))` whenever numerical input may contain decimal values.

5. Step 3: Understand the Conditional Logic

The program must choose one of three actions:

1. If the unit is `C`, convert Celsius to Fahrenheit.
2. If the unit is `F`, convert Fahrenheit to Celsius.
3. Otherwise, display an invalid-unit message.

We can implement these choices using `if`, `elif`, and `else`.

General syntax

```
if condition1:
    # Executes if condition1 is True
elif condition2:
    # Executes if condition1 is False
    # and condition2 is True
else:
    # Executes if neither condition matches
```

How it applies to this program

```
if unit == "C":
    # Convert Celsius to Fahrenheit
elif unit == "F":
    # Convert Fahrenheit to Celsius
else:
    # Display an error message
```

Condition	Action
`unit == "C"`	*Convert Celsius to Fahrenheit*
`unit == "F"`	*Convert Fahrenheit to Celsius*
Neither matches	*Display an invalid-unit message*

Remember: *Use* `==` *to compare values. A single* `=` *assigns a value to a variable.*

6. Step 4: Convert Celsius to Fahrenheit

If the user enters `C`*, the program should apply:*

$$F = \frac{9}{5}C + 32$$

Python code

```
if unit == "C":
    temp = (temp * 9 / 5) + 32
```

Suppose the temperature is $33^\circ C$.

```
temp = 33.0

temp = (temp * 9 / 5) + 32

print(temp)
```

Output:

```
91.4
```

The `temp` *variable is reassigned to store the converted temperature.*

Display the result

```
print(f"The temperature in Fahrenheit is {temp}°F")
```

7. Step 5: Convert Fahrenheit to Celsius

If the user enters `F`, the program should apply:

$$C = (F - 32) \times \frac{5}{9}$$

Python code

```
elif unit == "F":
    temp = (temp - 32) * 5 / 9
```

Suppose the temperature is $91.4^\circ F$.

```
temp = 91.4

temp = (temp - 32) * 5 / 9

print(temp)
```

Output:

```
33.00000000000001
```

You may occasionally see a very small floating-point difference like this because many decimal fractions cannot be represented exactly in binary floating-point arithmetic.

The value is effectively $33^\circ C$ for this calculation.

Display the result

```
print(f"The temperature in Celsius is {temp}°C")
```

The program will display a value close to:

```
The temperature in Celsius is 33.0°C
```

To make the output consistently show one decimal place, round the value.

8. Step 6: Round the Converted Temperature

Python's built-in `round()` function rounds a number to a specified number of decimal places.

Syntax

```
round(number, ndigits)
```

- `number`: the numeric value to round.
- `ndigits`: the number of decimal places to retain.

For this exercise, we want one decimal place.

Example 1: Round Fahrenheit

```
temp = 91.4444

temp = round(temp, 1)

print(temp)
```

9. Step 7: Handle Invalid Units

What happens if the user enters `K` for Kelvin or another unsupported value?

Neither the `if` nor the `elif` condition will match.

Use the `else` statement to handle this case.

```
if unit == "C":
    temp = (temp * 9 / 5) + 32

elif unit == "F":
    temp = (temp - 32) * 5 / 9

else:
    print(f"{unit} is an invalid unit of measurement.")
```

Example:

```
Is the temperature in Celsius or Fahrenheit? (C/F): K
Enter the temperature: 100
K is an invalid unit of measurement.
```

However, the program should not display a converted temperature after invalid input.

One straightforward approach is to print the conversion result inside each valid branch. Another option is to use a variable to track whether the conversion succeeded.

The complete program below uses the first approach to keep the structure beginner-friendly.

10. Complete Beginner Temperature Converter

This version follows the structure of the video. It accepts a unit, accepts a temperature, converts it, rounds the result to one decimal place, and handles invalid units.

```
# Python Temperature Converter

# Ask for the current temperature unit
unit = input("Is the temperature in Celsius or Fahrenheit? (C/F): ")

# Accept the temperature
temp = float(input("Enter the temperature: "))

# Convert Celsius to Fahrenheit
if unit == "C":
    temp = (temp * 9 / 5) + 32
    temp = round(temp, 1)

    print(f"The temperature in Fahrenheit is {temp:.1f}°F")

# Convert Fahrenheit to Celsius
elif unit == "F":
    temp = (temp - 32) * 5 / 9
    temp = round(temp, 1)

    print(f"The temperature in Celsius is {temp:.1f}°C")

# Handle invalid units
else:
    print(f"{unit} is an invalid unit of measurement.")
```

Sample execution 1: Celsius to Fahrenheit

```
Is the temperature in Celsius or Fahrenheit? (C/F): C
Enter the temperature: 33
The temperature in Fahrenheit is 91.4°F
```

Sample execution 2: Fahrenheit to Celsius

```
Is the temperature in Celsius or Fahrenheit? (C/F): F
Enter the temperature: 91.4
The temperature in Celsius is 33.0°C
```

Sample execution 3: Invalid unit

```
Is the temperature in Celsius or Fahrenheit? (C/F): K
Enter the temperature: 100
K is an invalid unit of measurement.
```

Notice that the invalid-unit case displays an error message instead of a converted temperature.

II. Improved Version: Accept Uppercase and Lowercase Units

The original program expects uppercase `C` *or* `F`*. We can improve it so that inputs such as* `c`*,* `f`*, or entries with surrounding spaces are also accepted.*

The `.strip()` *method removes whitespace from the beginning and end of a string, while* `.upper()` *converts letters to uppercase.*

```
# Improved Temperature Converter

unit = input(
    "Is the temperature in Celsius or Fahrenheit? (C/F): "
).strip().upper()

temp = float(input("Enter the temperature: "))

if unit == "C":
    converted_temp = (temp * 9 / 5) + 32
    print(f"The temperature in Fahrenheit is {converted_temp:.1f}°F")

elif unit == "F":
    converted_temp = (temp - 32) * 5 / 9
    print(f"The temperature in Celsius is {converted_temp:.1f}°C")

else:
    print(f"{unit} is an invalid unit of measurement.")
```

Why is this version better?

- `C`*,* `c`*, and* `" c "` *are all accepted as Celsius.*
- `F`*,* `f`*, and* `" f "` *are all accepted as Fahrenheit.*
- *A separate* `converted_temp` *variable preserves the original temperature.*
- *The f-string displays the result with exactly one decimal place.*

Note: *Both versions assume the user enters a valid number. If the user types text such as* `hot` *instead of a temperature,* `float()` *raises a* `ValueError`*. Input validation can be added later using* `try-except`*.*

12. Common Mistakes to Avoid

Mistake 1: Forgetting type casting

Incorrect:

```
temp = input("Enter the temperature: ")
temp = (temp * 9 / 5) + 32
```

The input is a string and cannot be used directly in this mathematical formula.

Correct:

```
temp = float(input("Enter the temperature: "))
```

Mistake 2: Reversing the formula

For Celsius to Fahrenheit, use:

```
fahrenheit = (celsius * 9 / 5) + 32
```

For Fahrenheit to Celsius, use:

```
celsius = (fahrenheit - 32) * 5 / 9
```

Do not use the same formula for both conversions.

Mistake 3: Forgetting parentheses

Incorrect:

```
celsius = fahrenheit - 32 * 5 / 9
```

This calculates `fahrenheit - (32 * 5 / 9)`*, which is not the required formula.*

Correct:

```
celsius = (fahrenheit - 32) * 5 / 9
```

Mistake 4: Forgetting the colon

Incorrect:

```
if unit == "C"
    print("Convert to Fahrenheit")
```

Correct:

```
if unit == "C":
    print("Convert to Fahrenheit")
```

Mistake 5: Incorrect indentation

Statements inside an `if`*,* `elif`*, or* `else` *block must be indented.*

```
if unit == "C":
    temp = (temp * 9 / 5) + 32
    print(f"{temp:.1f}°F")
```

13. Practice Questions

Try solving these questions before checking the answers.

A. Predict the output

Question 1

```
temp = 25
temp = (temp * 9 / 5) + 32
print(round(temp, 1))
```

Question 2

```
temp = 68
temp = (temp - 32) * 5 / 9
print(round(temp, 1))
```

Question 3

```
unit = "C"

if unit == "C":
    print("Celsius selected.")
elif unit == "F":
    print("Fahrenheit selected.")
else:
    print("Invalid unit.")
```

Question 4

```
unit = "K"

if unit == "C":
    print("Convert to Fahrenheit.")
elif unit == "F":
    print("Convert to Celsius.")
else:
    print("Invalid unit.")
```

Question 5

Why is `float(input( ... ))` used instead of simply `input( ... )` when accepting a temperature?

B. Conceptual questions

1. What is a temperature conversion program?
2. Write the formula for converting Celsius to Fahrenheit.
3. Write the formula for converting Fahrenheit to Celsius.
4. What is the purpose of `float()`?
5. What is the purpose of `round()`?
6. What is the difference between `if`, `elif`, and `else`?
7. Why are parentheses required in `(temp - 32) * 5 / 9`?
8. How can you allow both `c` and `C` as valid input?
9. What happens if the user enters an invalid unit?
10. What is the benefit of using `{temp:.1f}` in an f-string?

14. Mini Project Challenges

After completing the basic converter, try making these improvements yourself.

Challenge 1: Add lowercase support

Allow users to enter `c` or `f`, as well as uppercase letters.

Hint: Use `.strip().upper()`.

Challenge 2: Display two decimal places

Change the result format from one decimal place to two.

Hint:

```
print(f"Temperature: {converted_temp:.2f}°C")
```

Challenge 3: Add input validation

Prevent the program from crashing if the user enters `hot` or another invalid numerical value.

Hint: Use `try-except` to catch `ValueError`.

Challenge 4: Add Kelvin support

Extend the program to convert Celsius to Kelvin and Kelvin to Celsius.

The formulas are:

$$K = C + 273.15$$

$$C = K - 273.15$$

Note: Kelvin is written as `K`, not `°K`.

15. Last-Minute Revision Sheet

Important formulas

Conversion	Formula
Celsius → Fahrenheit	`(temp * 9 / 5) + 32`
Fahrenheit → Celsius	`(temp - 32) * 5 / 9`

Essential Python concepts

Concept	Purpose
`input()`	Accepts the unit and temperature.
`float()`	Converts temperature input into a numeric value.
`if`	Checks whether the unit is Celsius.
`elif`	Checks whether the unit is Fahrenheit.
`else`	Handles unsupported units.
`round()`	Rounds the converted temperature.
`.upper()`	Converts unit input to uppercase.
`.strip()`	Removes surrounding whitespace.
F-string	Displays the result with its unit.

Five key points to remember

1. Use `float(input( ... ))` to accept numeric temperatures.
2. Multiply by $9/5$ and add 32 to convert Celsius to Fahrenheit.
3. Subtract 32 and multiply by $5/9$ to convert Fahrenheit to Celsius.
4. Use `if-elif-else` to select the appropriate conversion.
5. Use `{converted_temp:.1f}` when you want the displayed result to have exactly one decimal place.

FINAL TAKEAWAY

The temperature converter is an excellent beginner project because it combines variables, user input, type casting, arithmetic formulas, conditional statements, and formatted output in one program.

Practice tip: Write the program from memory, test it with `C`, `F`, and an invalid unit, and then extend it to accept lowercase input and Kelvin. This will reinforce your understanding of Python decision-making and basic mathematical programming.